Drayan Mathematical Series

Classical Algebra

Problem

Solve the system

$$\begin{aligned} x + y + z &= a, \\ px + qy + rz &= b, \\ p^2x + q^2y + r^2z &= c, \\ p^3x + q^3y + r^3z &= d, \\ p^4x + q^4y + r^4z &= e, \\ p^5x + q^5y + r^5z &= f, \end{aligned}$$

where x, y, z, p, q, r are unknowns.
Determine the solution when

$$a = 2, \qquad b = 3, \qquad c = 4, \qquad d = 6, \qquad e = 12, \qquad f = 32.$$

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S.Ramanujan

Solution

Define

$$V_n = p^n x + q^n y + r^n z.$$

From the given data,

$$V_0 = 2, \quad V_1 = 3, \quad V_2 = 4, \quad V_3 = 6, \quad V_4 = 12, \quad V_5 = 32.$$

Let p, q, r be the roots of

$$t^3 - S_1t^2 + S_2t - S_3 = 0,$$

where

$$S_1 = p + q + r, \qquad S_2 = pq + pr + qr, \qquad S_3 = pqr.$$

Since every root satisfies the characteristic equation,

$$t^{n+3} = S_1t^{n+2} - S_2t^{n+1} + S_3t^n.$$

Multiplying by x, y, z respectively and summing yields

$$V_{n+3} - S_1V_{n+2} + S_2V_{n+1} - S_3V_n = 0.$$

Applying this recurrence for $n = 0, 1, 2$, we obtain

$$4S_1 - 3S_2 + 2S_3 = 6,$$

$$6S_1 - 4S_2 + 3S_3 = 12,$$

$$12S_1 - 6S_2 + 4S_3 = 32.$$

Dividing the third equation by 2,

$$6S_1 - 3S_2 + 2S_3 = 16.$$

Subtracting the first equation gives

$$2S_1 = 10,$$

hence

$$S_1 = 5.$$

Substituting into the first two equations,

$$3S_2 - 2S_3 = 14,$$

$$4S_2 - 3S_3 = 18.$$

Multiplying by 3 and 2 respectively,

$$9S_2 - 6S_3 = 42,$$

$$8S_2 - 6S_3 = 36.$$

Subtracting,

$$S_2 = 6.$$

Substituting back,

$$18 - 2S_3 = 14,$$

so

$$S_3 = 2.$$

Therefore

$$p, q, r$$

are roots of

$$t^3 - 5t^2 + 6t - 2 = 0.$$

Since

$$1 - 5 + 6 - 2 = 0,$$

$t = 1$ is a root and

$$t^3 - 5t^2 + 6t - 2 = (t-1)(t^2 - 4t + 2).$$

The remaining roots are

$$t = \frac{4 \pm \sqrt{8}}{2} = 2 \pm \sqrt{2}.$$

Thus, up to permutation,

$$p = 1, \qquad q = 2 + \sqrt{2}, \qquad r = 2 - \sqrt{2}.$$

We now solve

$$x + y + z = 2,$$

$$x + (2+\sqrt{2})y + (2-\sqrt{2})z = 3,$$

$$x + (6+4\sqrt{2})y + (6-4\sqrt{2})z = 4.$$

Subtracting the first equation from the latter two gives

$$(1+\sqrt{2})y + (1-\sqrt{2})z = 1,$$

$$(5+4\sqrt{2})y + (5-4\sqrt{2})z = 2.$$

Multiplying the first equation by 4 and subtracting,

$$y + z = -2.$$

Hence

$$z = -2 - y.$$

Substituting into the first equation,

$$(1+\sqrt{2})y + (1-\sqrt{2})(-2-y) = 1.$$

After simplification,

$$2\sqrt{2}\,y = 3 - 2\sqrt{2},$$

so

$$y = \frac{3\sqrt{2}}{4} - 1.$$

Therefore

$$z = -1 - \frac{3\sqrt{2}}{4}.$$

Finally,

$$x = 2 - y - z = 4.$$

Answer

$$\boxed{\begin{aligned} &p = 1, \\ &q = 2 + \sqrt{2}, \\ &r = 2 - \sqrt{2}, \\ &x = 4, \\ &y = \frac{3\sqrt{2} - 4}{4}, \\ &z = -\frac{3\sqrt{2} + 4}{4}. \end{aligned}}$$